
Test Summary Report

Project: Apex – Real-Time Data Integrity Framework for a Financial Charting Platform

Application Under Test (AUT): TradingView Advanced Chart (UI)

Source-of-Truth API: Alpha Vantage API

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1. Introduction

The purpose of this Test Summary Report is to provide a detailed overview of the testing activities carried out for the **Apex Framework** project. The focus of testing was to validate **real-time data integrity** between the TradingView Advanced Chart UI and Alpha Vantage API, ensure correct chart interactions, and confirm real-time liveness of data streams.

2. Objectives

- Validate **Historical Data Integrity** (UI OHLC vs. API OHLC).

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- Verify complex **Chart Interactions** (Indicators, Time Frame changes).
 - Confirm **Real-Time Price Updates** and formatting.
 - Ensure framework stability under **CI/CD pipeline execution**.
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3. Scope of Testing

In Scope:

- UI testing of TradingView Advanced Chart (web).
- API testing with Alpha Vantage.
- Data validation of OHLC values.
- Real-time data liveness checks.
- Reporting and CI/CD integration.

Out of Scope:

- Performance testing (load/stress).
 - Security and penetration testing.
 - Mobile application testing.
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4. Test Environment

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- **Browser:** Chrome (latest), Headless mode for CI/CD.
 - **Framework:** Selenium + TestNG + Cucumber + Extent Reports.
 - **CI/CD:** GitHub Actions (Linux runner).
 - **API:** Alpha Vantage (Free API Key).
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5. Test Scenarios Executed

Module	Scenario	Status
Data Integrity	Compare OHLC values (UI vs API) for AAPL	✓ Passed
Chart Interactions	Add the MACD indicator and validate	✓ Passed
Chart Interactions	Change time frame (1D → 1W → 1M)	✓ Passed
Real-Time Validation	Validate ticker liveness (5s gap)	✓ Passed
Real-Time Validation	Validate ticker format with Regex	✓ Passed

6. Defect Summary

Defect ID	Summary	Severity	Status
BUG-01	Minor mismatch in OHLC values due to latency (UI vs API)	Medium	Open
BUG-02	Delay in rendering the MACD panel under a slow network	Low	Closed
BUG-03	Time frame update lag (UI freeze 2–3s)	Medium	Open

7. Key Findings

- UI data was found to be **99% consistent** with API data; minor mismatches were attributed to **real-time latency**.
- Real-time ticker validation confirmed **continuous live updates** with proper formatting.
- The framework successfully automated **complex chart interactions** using the Actions class.
- CI/CD pipeline executed all tests in **headless mode without failure**.

8. Conclusion

The Apex Framework has successfully demonstrated its capability to validate **data integrity** between a real-time charting UI and an authoritative financial API. The

framework proved robust, reliable, and automation-ready, with detailed reporting via Extent Reports and smooth CI/CD integration.

9. Recommendations

- Enhance synchronization strategies to reduce false mismatches caused by latency.
- Extend coverage to performance and security testing in future phases.
- Consider integrating multiple financial APIs for redundancy.